Owen Skriloff

Contact Information

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Research Interests

I want to understand how prior scientific knowledge shapes machine learning architecture choices. My approach uses geometric, statistical, and algebraic tools to probe the latent representations of models. By studying how constraints influence both the training process and the resulting latent spaces, I aim to develop systematic methods for designing intelligent systems that fully leverage existing scientific knowledge. I am particularly interested in the application of physics-informed machine learning for chemical and biological problems and the application of tools from statistical physics to data science.

Education

MS, Tufts University School of Arts and Sciences (Expected) May 2026
Mathematics
GPA: 4.0, Concentration in Foundations of Machine Learning

BS, Tufts University School of Engineering May 2025
Chemical and Biological Engineering and Mathematics
GPA: 3.9, Highest Thesis Honors, Summa Cum Laude

Awards

Benjamin G Brown Scholarship *(Research Excellence)* 2025
Christos Georgakis Award *(Best Thesis)* 2025
Chemical Engineering Summer Research Grant 2024
Regeneron Science Talent Search Top Scholar 2020
National Science and Humanities Symposium: 3rd Place Physical Sciences 2020

Publications

[1] O. Skriloff, D. Stoukides, E. Tzanakakis. *Augmentation models of stem cell culture data for machine learning. Under Review*, 2025.

Conferences

[1] O. Skriloff, D. Stoukides, E. Tzanakakis. *Data Augmentation Models for Machine Learning Applications in Stem Cell Bioprocessing.* AIChE Annual Meeting, 2025. **Speaker.**

[2] O. Skriloff, D. Stoukides, E. Tzanakakis. *Data augmentation models for stem cell bioprocessing.* Cell Culture Engineering XIX (ECI). **Poster.**

Employment

Tzanakakis Research Group Medford, MA
Researcher *Jan 2023 - May 2025*
- Engineered and trained three physics-informed ML models of stem cell growth dynamics.
- Built novel data pipeline with Monte Carlo simulation and feature importance to quantify and visualize representational alignment between models and known biological behavior.
- Determined optimal model class and design procedure for stem cell manufacturing.

Itaconix Corporation Stratham, NH
Summer Associate Engineer *Jun 2023 - Aug 2023*
- Commissioned, purchased parts for, and assembled a 100L SS reactor with vacuum pump.
- Scaled an esterification process from 250mL to the first pilot production at 50L.
- Developed HPLC and GPC methods for 200+ samples and supported GPC installation.

Araca Inc. Tucson, AZ
Laboratory Technician *Sep 2020 - Dec 2020*
- Conducted 25+ chemical mechanical planarization (CMP) and post-CMP cleaning experiments in a Class 100 cleanroom to optimize CMP conditions for semiconductor manufacturing.
- Measured 250+ wafers, mixed slurries, programmed polishers, and tested new CMP products.
- Analyzed data in MATLAB, Excel, and LabVIEW to prepare 25+ presentations visualizing experimental results to inform client process improvements.

Temple University Smart Biomaterials Lab Philadelphia, PA
Researcher *Mar 2019 - Aug 2020*

- Developed and performed methodology to quantify a relationship between stress and mineral deposits generated by electric charges from a piezoelectric composite using simulated body fluid, incubator, and actuator.
- Authored research report: An *in-vitro* evaluation of the relationship between stress and mineralization through the use of a piezoelectric barium titanate composite.

Selected Projects

Loss-landscape geometry and generalization across MLP depths
Final Project: Iterative Methods in ML *May 2025*

- Investigated how network depth affects generalization by analyzing typical loss landscape curvature in both Euclidean and Fisher-Rao geometry with random training restarts.
- Improved alignment of training with validation loss by 15% using geometric regularization.
- Discovered empirical broken scaling law ($R^2 = 0.93$) between loss landscape curvature and validation loss, with scaling break (critical point) at optimal depth.

Industrial amikacin production plant: process design and economic analysis
CHBE Capstone Project: With David Gillingham, Ivy Le, Youngseo Yi *May 2025*

- Designed and simulated industrial scale amikacin synthesis process including bioprocessing and chemical reaction steps using SuperPro, Excel, and Python.
- Performed Class IV economic analysis of profitability via labor, materials, and equipment costs.

Energy and Entropy Measures for Reaction Diffusion PDE Analysis
Final Project: Numerical PDEs *Dec 2024*

- Designed, proved stability and convergence, and implemented efficient finite element numerical solver in Python for reaction diffusion systems.
- Analyzed efficacy of entropy and energy measures for detecting emergent dynamic behavior.

Applicable Courses

Mathematics
Numerical PDEs, Iterative Methods in Machine Learning, Statistics, High Dimensional Probability, Mathematical Aspects of Data, Mathematical Modeling, Analysis, Algebra, Differential Geometry.

Chemistry and Chemical Engineering
Process Dynamics & Control, Transport Phenomena, Thermodynamics, Chemical Physics, Intro to Modern Physics, Polymerization, Materials Science, Numerical Methods, Organic Chemistry.

Skills

Computer Languages
Python, MATLAB, LaTeX, HTML.

Scientific Computing
Neural networks (PyTorch, TensorFlow), data science (scikit-learn, NumPy, pandas), data visualization (seaborn, matplotlib), version control (Git, Unix, Bash), object-oriented programming, Monte Carlo simulation, optimization, finite element solvers.

Engineering
Reaction design and scale-up, ChemStation, HPLC and GPC, chemical mechanical planarization, reaction kinetic modeling, process control modeling, SuperPro Designer.

Personal Interests

Outside of academics, I'm an avid reader and artist. I love to draw, paint, play saxophone, and spend time traveling and in museums. I also enjoy skiing, running, hiking, and doing flips.